# American Computer Science League

2020-2021 ● Contest 4: Shorts ● Classroom Division

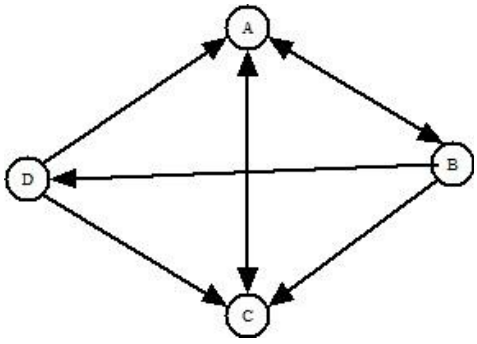
**1. Graph Theory**

Given the set of vertices: {A, B, C, D, E} and the set of directed edges: {AB, BE, AD, EA, CB, DB, BA, BC}, how many paths are there of length 2 from vertex B?

A. 1
B. 2
C. 3
D. 4
E. 5

**2. Graph Theory**

Given the following directed graph, how many cycles are there from A?




A. 7
B. 6
C. 5
D. 4
E. 3

**3. Graph Theory**

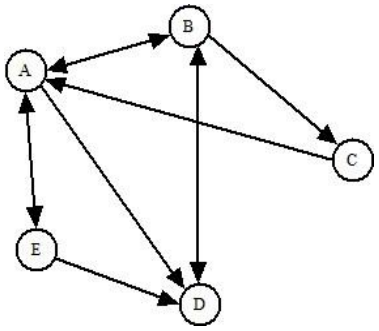
Given the following adjacency matrix, which vertices do not have a path of length 3 to vertex B?

| | A | B | C | D |
|---|---|---|---|---|
| A | 1 | 0 | 0 | 1 |
| B | 0 | 1 | 1 | 0 |
| C | 1 | 0 | 1 | 0 |
| D | 0 | 0 | 1 | 1 |

A. A, C, D
B. A, B, C
C. A, B, D
D. C, D
E. A, C

**4. Graph Theory**

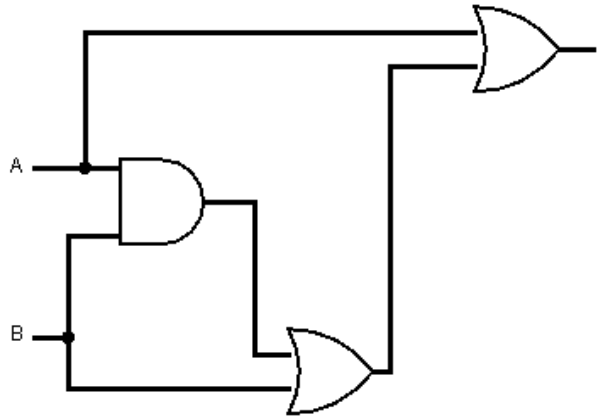
How many cycles are there in this directed graph?




A. 5
B. 6
C. 7
D. 8
E. 9

**5. Digital Electronics**

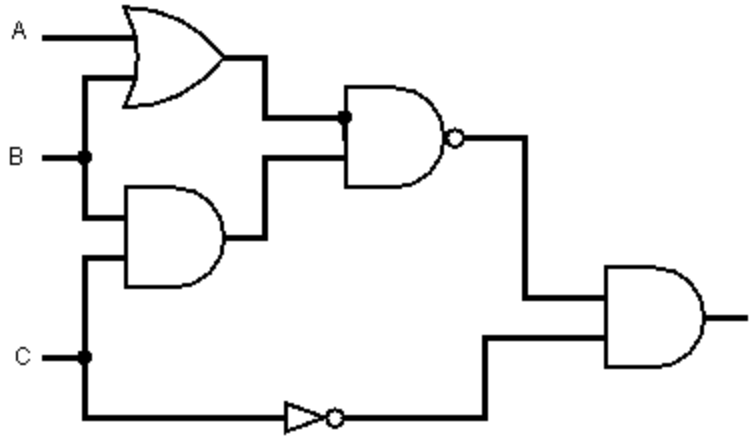
Simplify the Boolean expression represented by the following digital circuit:




A. A
B. A+B
C. B
D. AB
E. 1

**6. Digital Electronics**

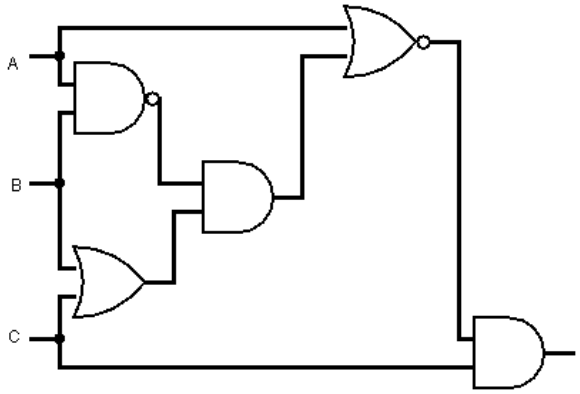
Identify all possible ordered triples that make the digital circuit TRUE.




A. (*,*,0)
B. (*,0,1)
C. (*,*,1)
D. (0,*,1)
E. (0,*,*)

**7. Digital Electronics**

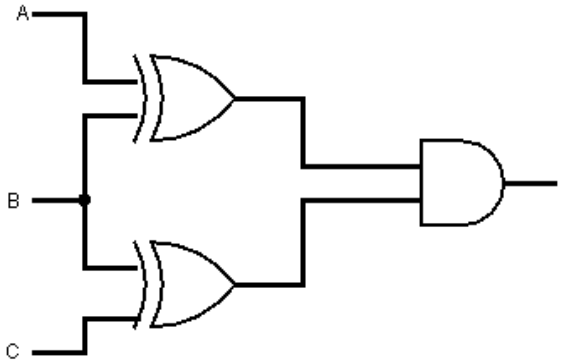
How many ordered triples make this circuit TRUE?




A. 0
B. 1
C. 2
D. 3
E. 4

**8. Digital Electronics**

Identify the ordered triples that make this circuit TRUE:




A. (0,*,0)
B. (1,*,1)
C. (0,*,0), (1,0,1)
D. (1,*,1), (0,1,0)
E. (0,1,0), (1,0,1)

**9. Assembly Language**

What is printed when the following program is executed?

```
N     DC      47
      LOAD    N
      DIV     =10
      STORE   P
      MULT    =10
      SUB     N
      MULT    =-1
      ADD     P
      STORE   S
      PRINT   S
      END
```

A. 4
B. 7
C. 11
D. 28
E. 47

**10. Assembly Language**

What is printed when the following program is executed?

```
X     DC      21
Y     DC      49
      LOAD    Y
TOP   BE      STOP
      LOAD    X
      DIV     Y
      MULT    Y
      STORE   R
      LOAD    X
      SUB     R
      STORE   R
      LOAD    Y
      STORE   X
      LOAD    R
      STORE   Y
      BU      TOP
STOP  PRINT   X
      END
```

A. 1
B. 7
C. 14
D. 21
E. 49